$C_{\phi u}^{(22)}$ i1_i2_ $\to$

$$\frac{1}{16\pi^2}\Big(\frac{4}{81}\sum_p g_1^4\, \mathrm{LF}_{3,0}\big[m_{\tilde d}{}^{p}\big]\,\delta_{i1i2}-\frac{5}{54}\sum_p g_1^4\, \mathrm{LF}_{4,-1}\big[m_{\tilde d}{}^{p}\big]\,\delta_{i1i2}+\frac{16}{405}\sum_p g_1^4\, \mathrm{LF}_{5,-2}\big[m_{\tilde d}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{4}{27}\sum_p g_1^4\, \mathrm{LF}_{3,0}\big[m_{\tilde e}{}^{p}\big]\,\delta_{i1i2}-\frac{5}{18}\sum_p g_1^4\, \mathrm{LF}_{4,-1}\big[m_{\tilde e}{}^{p}\big]\,\delta_{i1i2}+\frac{16}{135}\sum_p g_1^4\, \mathrm{LF}_{5,-2}\big[m_{\tilde e}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{2}{27}\sum_p g_1^4\, \mathrm{LF}_{3,0}\big[m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}-\frac{5}{36}\sum_p g_1^4\, \mathrm{LF}_{4,-1}\big[m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}+\frac{8}{135}\sum_p g_1^4\, \mathrm{LF}_{5,-2}\big[m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{2}{81}\sum_p g_1^4\, \mathrm{LF}_{3,0}\big[m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}-\frac{5}{108}\sum_p g_1^4\, \mathrm{LF}_{4,-1}\big[m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}+\frac{8}{405}\sum_p g_1^4\, \mathrm{LF}_{5,-2}\big[m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{16}{81}\sum_p g_1^4\, \mathrm{LF}_{3,0}\big[m_{\tilde u}{}^{p}\big]\,\delta_{i1i2}-\frac{10}{27}\sum_p g_1^4\, \mathrm{LF}_{4,-1}\big[m_{\tilde u}{}^{p}\big]\,\delta_{i1i2}+\frac{64}{405}\sum_p g_1^4\, \mathrm{LF}_{5,-2}\big[m_{\tilde u}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{2}{27}g_1^4\,\mathrm{LF}_{3,0}[\tilde\mu]\,\delta_{i1i2}+\frac{1}{9}g_1^4\,\mathrm{LF}_{4,-1}[\tilde\mu]\,\delta_{i1i2}-\frac{16}{135}g_1^4\,\mathrm{LF}_{5,-2}[\tilde\mu]\,\delta_{i1i2}+$$

$$\frac{2}{81}g_1^4\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+\frac{2}{81}g_1^4\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}-\frac{4}{81}g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+$$

$$\frac{2}{81}g_1^4\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+\frac{2}{81}g_1^4\,\mathrm{LF}_{2,1,0}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+\frac{2}{81}g_1^4\,\mathrm{LF}_{2,2,-1}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}-$$

$$\frac{4}{81}g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+\frac{2}{81}g_1^4\,\mathrm{LF}_{4,1,-2}\big[m_1,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}-$$

$$\frac{4}{27}g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{2,2,-1}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}-$$

$$\frac{4}{27}g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{4,1,-2}\big[m_3,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+$$

$$\frac{2}{9}g_1^2\,\overline{a_d}^{pr}\,a_d{}^{pr}\,\mathrm{LF}_{2,2,0}\big[m_{\tilde d}{}^{r},m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}-\frac{1}{9}g_1^2\,\overline{a_d}^{pr}\,a_d{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde d}{}^{r},m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{2}{9}g_1^2\,\overline{a_e}^{pr}\,a_e{}^{pr}\,\mathrm{LF}_{2,2,0}\big[m_{\tilde e}{}^{r},m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}-\frac{1}{9}g_1^2\,\overline{a_e}^{pr}\,a_e{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde e}{}^{r},m_{\tilde l}{}^{p}\big]\,\delta_{i1i2}-$$

$$\frac{2}{9}g_1^2\,\overline{a_e}^{pr}\,a_e{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde l}{}^{p},m_{\tilde e}{}^{r}\big]\,\delta_{i1i2}-\frac{2}{9}g_1^2\,\overline{a_e}^{pr}\,a_e{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde l}{}^{p},m_{\tilde e}{}^{r}\big]\,\delta_{i1i2}+$$

$$\frac{1}{2}g_1^2\,\overline{a_e}^{pr}\,a_e{}^{pr}\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde l}{}^{p},m_{\tilde e}{}^{r}\big]\,\delta_{i1i2}-\frac{2}{9}g_1^2\,\overline{a_e}^{pr}\,a_e{}^{pr}\,\mathrm{LF}_{5,1,-2}\big[m_{\tilde l}{}^{p},m_{\tilde e}{}^{r}\big]\,\delta_{i1i2}-$$

$$\frac{2}{9}g_1^2\,\overline{a_d}^{pr}\,a_d{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde q}{}^{p},m_{\tilde d}{}^{r}\big]\,\delta_{i1i2}-\frac{2}{9}g_1^2\,\overline{a_d}^{pr}\,a_d{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde q}{}^{p},m_{\tilde d}{}^{r}\big]\,\delta_{i1i2}+$$

$$\frac{5}{6}g_1^2\,\overline{a_d}^{pr}\,a_d{}^{pr}\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde q}{}^{p},m_{\tilde d}{}^{r}\big]\,\delta_{i1i2}-\frac{2}{3}g_1^2\,\overline{a_d}^{pr}\,a_d{}^{pr}\,\mathrm{LF}_{5,1,-2}\big[m_{\tilde q}{}^{p},m_{\tilde d}{}^{r}\big]\,\delta_{i1i2}-$$

$$\frac{1}{9}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\,\mathrm{LF}_{2,2,0}\big[m_{\tilde q}{}^{p},m_{\tilde u}{}^{r}\big]\,\delta_{i1i2}+\frac{1}{18}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde q}{}^{p},m_{\tilde u}{}^{r}\big]\,\delta_{i1i2}-$$

$$\frac{1}{18}g_1^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,0}\big[m_{\tilde q}{}^{p},\tilde\mu\big]+\frac{1}{36}g_1^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,2,-1}\big[m_{\tilde q}{}^{p},\tilde\mu\big]+$$

$$\frac{1}{36}g_1^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde q}{}^{p},\tilde\mu\big]+\frac{1}{9}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\,\mathrm{LF}_{3,1,0}\big[m_{\tilde u}{}^{r},m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}+$$

$$\frac{1}{9}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\,\mathrm{LF}_{3,2,-1}\big[m_{\tilde u}{}^{r},m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}+\frac{1}{3}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\,\mathrm{LF}_{4,1,-1}\big[m_{\tilde u}{}^{r},m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}-$$

$$\frac{2}{3}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u{}^{pr}\,\mathrm{LF}_{5,1,-2}\big[m_{\tilde u}{}^{r},m_{\tilde q}{}^{p}\big]\,\delta_{i1i2}-\frac{4}{81}g_1^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i1},m_1\big]\,\delta_{i1i2}+$$

$$\frac{2}{81}g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i1},m_1\big]\,\delta_{i1i2}-\frac{4}{27}g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i1},m_3\big]\,\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i1},m_3\big]\,\delta_{i1i2}-\frac{4}{81}g_1^4\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i2},m_1\big]\,\delta_{i1i2}+$$

$$\frac{2}{81}g_1^4\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i2},m_1\big]\,\delta_{i1i2}-\frac{4}{27}g_1^2\,g_3^2\,\mathrm{LF}_{2,1,0}\big[m_{\tilde u}{}^{i2},m_3\big]\,\delta_{i1i2}+$$

$$\frac{2}{27}g_1^2\,g_3^2\,\mathrm{LF}_{3,1,-1}\big[m_{\tilde u}{}^{i2},m_3\big]\,\delta_{i1i2}-\frac{5}{18}g_1^4\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_1]\,\delta_{i1i2}+$$

$$\frac{2}{9}g_1^4\,\mathrm{LF}_{5,1,-3}[\tilde\mu,m_1]\,\delta_{i1i2}-\frac{5}{6}g_1^2\,g_2^2\,\mathrm{LF}_{4,1,-2}[\tilde\mu,m_2]\,\delta_{i1i2}+\frac{2}{3}g_1^2\,g_2^2\,\mathrm{LF}_{5,1,-3}[\tilde\mu,m_2]\,\delta_{i1i2}+$$

$$\frac{1}{36}g_1^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,0}\big[\tilde\mu,m_{\tilde q}{}^{p}\big]-\frac{11}{36}g_1^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{3,1,-1}\big[\tilde\mu,m_{\tilde q}{}^{p}\big]+$$

$$\frac{1}{9}g_1^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{4,1,-2}\big[\tilde\mu,m_{\tilde q}{}^{p}\big]-\frac{1}{18}g_1^4\,\mathrm{LF}_{2,1,1,-1}\big[m_1,m_{\tilde u}{}^{i1},\tilde\mu\big]\,\delta_{i1i2}-$$

$$\frac{1}{9}g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}\big[m_1,m_{\tilde u}{}^{i1},\tilde\mu\big]\,\delta_{i1i2}-\frac{1}{18}g_1^4\,\mathrm{LF}_{2,1,1,-1}\big[m_1,m_{\tilde u}{}^{i2},\tilde\mu\big]\,\delta_{i1i2}-$$

$$\frac{1}{9}g_1^4\,m_1^2\,\mathrm{LF}_{2,1,1,0}\big[m_1,m_{\tilde u}{}^{i2},\tilde\mu\big]\,\delta_{i1i2}+\frac{1}{8}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,2,1,-1}\big[m_1,\tilde\mu,m_{\tilde q}{}^{p}\big]-$$

$$\frac{1}{18}g_1^4\,\mathrm{LF}_{2,2,1,-2}\big[m_1,\tilde\mu,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}+\frac{1}{18}g_1^4\,m_1^2\,\mathrm{LF}_{2,2,1,-1}\big[m_1,\tilde\mu,m_{\tilde u}{}^{i1}\big]\,\delta_{i1i2}-$$

$$\frac{1}{18}g_1^4\,\mathrm{LF}_{2,2,1,-2}\big[m_1,\tilde\mu,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+\frac{1}{18}g_1^4\,m_1^2\,\mathrm{LF}_{2,2,1,-1}\big[m_1,\tilde\mu,m_{\tilde u}{}^{i2}\big]\,\delta_{i1i2}+$$

$$\frac{3}{8}g_2^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,2,1,-1}\big[m_2,\tilde\mu,m_{\tilde q}{}^{p}\big]-\frac{1}{4}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,0}\big[\tilde\mu,m_1,m_{\tilde q}{}^{p}\big]-$$

$$\frac{3}{4}g_2^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,0}\big[\tilde\mu,m_2,m_{\tilde q}{}^{p}\big]-\frac{1}{6}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{1,1,1,1,0}\big[m_1,m_{\tilde q}{}^{p},m_{\tilde u}{}^{i1},\tilde\mu\big]+$$

$$\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_1,m_{\tilde q}{}^{p},m_{\tilde u}{}^{i1},\tilde\mu\big]-$$

$$\frac{1}{6}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{1,1,1,1,0}\big[m_1,m_{\tilde q}{}^{p},m_{\tilde u}{}^{i2},\tilde\mu\big]+\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}$$

$$\mathrm{LF}_{2,1,1,1,-1}\big[m_1,m_{\tilde q}{}^{p},m_{\tilde u}{}^{i2},\tilde\mu\big]-\frac{1}{4}\overline{y_u}^{si1}\,y_u{}^{pi2}\,\overline{a_d}^{pr}\,a_d{}^{sr}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde d}{}^{r},m_{\tilde q}{}^{p},m_{\tilde q}{}^{s},\tilde\mu\big]-$$

$$\frac{2}{9}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde q}{}^{p},m_1,m_{\tilde u}{}^{i1},m_{\tilde u}{}^{i2}\big]-\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}$$

$$\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde q}{}^{p},m_1,m_{\tilde u}{}^{i1},\tilde\mu\big]-\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde q}{}^{p},m_1,m_{\tilde u}{}^{i2},\tilde\mu\big]-$$

$$\frac{2}{3}g_3^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde q}{}^{p},m_3,m_{\tilde u}{}^{i1},m_{\tilde u}{}^{i2}\big]+\frac{1}{4}\tilde\mu^2\,\overline{y_u}^{ps}\,\overline{y_u}^{ri1}\,y_u{}^{pi2}\,y_u{}^{rs}$$

$$\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde u}{}^{s},m_{\tilde q}{}^{p},m_{\tilde q}{}^{r},\tilde\mu\big]+\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde u}{}^{i1},m_1,m_{\tilde q}{}^{p},\tilde\mu\big]+$$

$$\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[m_{\tilde u}{}^{i2},m_1,m_{\tilde q}{}^{p},\tilde\mu\big]-\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}$$

$$\mathrm{LF}_{2,1,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde q}{}^{p},m_{\tilde u}{}^{i1}\big]-\frac{1}{12}g_1^2\,\tilde\mu^2\,\overline{y_u}^{pi1}\,y_u{}^{pi2}\,\mathrm{LF}_{2,1,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde q}{}^{p},m_{\tilde u}{}^{i2}\big]\Big)$$